

Instrument

$\{\mathbf{I}_L, \mathbf{I}_R\}$
stereo, 10-30 Hz

$\tilde{\mathbf{f}}_I, \tilde{\omega}_I$
scope IMU, 200 Hz

$B, \tilde{\omega}_B$
board witness

Optical reference

$\mathbf{T}_{BC} \leftarrow \text{PnP}(\mathbf{I}_L)$
 $e_{px} \leq 1.5, \|\tilde{\omega}_B\| < 8^\circ/\text{s}$

Inertial path

$\mathbf{R}_{a \leftarrow k} = \mathbf{R}_{CI} \mathbf{Q}_a^\top \mathbf{Q}_k \mathbf{R}_{CI}^\top$
 $\ell_k = \mathbf{R}_{a \leftarrow k} \mathbf{R}_{CI} \tilde{\mathbf{f}}_{I,k} - \hat{\boldsymbol{\mu}}_f$

What each path can supply

$$\Delta \mathbf{p}_{ab} = \underbrace{\mathbf{v}(t_a) \Delta t}_{\text{unobservable prior or pause}} + \underbrace{\int \mathbf{a} d\tau ds}_{\substack{\text{measured} \\ \pm \mathbf{b}, \pm s, \pm \delta\theta \times \mathbf{g}}}$$

Digital twin

$\Sigma, \mathbf{b} \sim \text{RW}, Q, \delta t, \mathbf{g}_B$
+ real-trajectory replay
 $\rightarrow \{\tilde{\mathbf{f}}, \tilde{\omega}\}_{\text{sim}}$

PhysNet

$\hat{\mathbf{v}}_k [1 - \sigma(s_k)]$
 $\hat{\mathbf{d}}_k = f_s^{-1} \sum \hat{\mathbf{v}} + (\mathbf{R} - \mathbf{I}) \lambda$
 $\rightarrow \mathbf{d}(t_b) u(\hat{\mathbf{d}})$

Audits

$u_A, \Sigma_{\text{corner}}$
 $s_p = \|\delta \mathbf{p}\| / \|\delta \theta\|$
 $r_{\text{axis}} \geq 0.75$
ideal/noisy, corners

Evaluation

e_p, R^2, ATE
synthetic / OOD / direct
8-fold session CV
held-out test, 95% CI

green: optical orange: inertial purple: simulation blue: estimation
red: nothing crosses back from the held-out evaluation